Automated Irregular Shaped Object Stacking via Pose Optimization and Monte Carlo Tree Search

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***Abstract*—This paper presents a novel approach for automated stacking of irregularly shaped objects. The proposed method combines Monte Carlo Tree Search (MCTS) with optimization techniques to achieve stable and efficient stacking configurations.**

I. Introduction

Automating the process of stacking objects has long been studied in the context of packing and warehousing systems. Our contributions are as follows:

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The rest of the paper is organized as follows. Sec. II Sec. III Sec. IV Sec. V

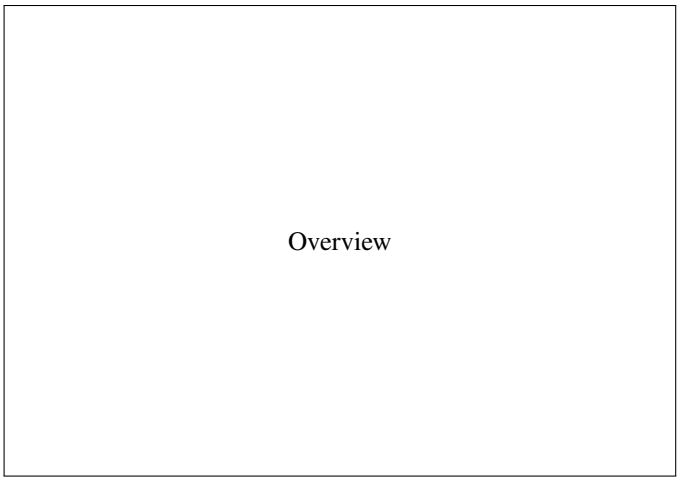


Fig. 1: Overview.

II. Related Works

III. Overview

A. Stacking as an Optimization Problem

Suppose that we are given a target shape $T \subset \mathbb{R}^3$ and n irregular shaped objects $O_1(q_1), O_2(q_2), \cdots, O_n(q_n) \subset \mathbb{R}^3$ where each $q_i \in \mathrm{SE}(3)$ represents the pose of each object O_i in the world frame. The goal is to find optimal placement poses $q_1, \cdots, q_n$ for a set of irregularly shaped objects such that the final configuration approximates the target shape T while satisfying the stability and non-collision constraints. This problem can be formulated as an optimization problem:

$$\min_{q_1,\cdots,q_n \in \mathrm{SE}(3)} \mathrm{v}\left(T \backslash \bigcup_{i=1}^{n} O_i(q_i)\right) \quad \text{s.t.} \quad c_{\text{phys}}(O_1(q_1), \cdots, O_n(q_n)) = 0 \tag{1}$$

where $\mathrm{v}(\cdot)$ is the volume of a 3D shape and $c_{\text{phys}}(\cdot)$ is a function that evaluates the physical feasibility of the configuration, including stability and non-collision constraints.

B. Shape Representation

Throughout the paper, we use the union of differentiable support function (DSF) to represent the 3D shapes of the target shape T and the objects O_i, $i \in \mathcal{I}_k$, in the inventory. Given a compact convex body $C \subset \mathbb{R}^3$, the support function $h_C : \mathbb{R}^3 \backslash \{0\} \to \mathbb{R}$ is defined as

$$h_C(x) = \max_{s \in C} x \cdot s$$

where $\cdot$ denotes the standard inner product in $\mathbb{R}^3$. The work of Lee et al. [8] presented a differentiable version of the support function h_C that represents a smooth shape C.

The function $h_C(x)$ can be easily extended through rigid-body transformation as follows:

$$h_{\bar{C}(q)}(x) = h_C(R^\top x) + t \cdot x$$

for $q := (R, t) \in \mathrm{SE}(3)$ where $\bar{C}(q) = \{Rp + t \mid p \in C\}$ denotes a transformed C under rigid-body transformation. Given two DSF represented convex bodies $C_1(q_1)$ and $C_2(q_2)$ with SE(3) poses q_1, q_2, the signed distance $\mathrm{d}(C_1(q_1), C_2(q_2))$ between the two convex bodies can be easily computed using Riemannian optimization techniques associated with $h_{C_1(q_1)}$ and $h_{C_2(q_2)}$ as presented in An, et al. [2]. Moreover, due to the smoothness of DSFs, the signed distance $\mathrm{d}(C_1(q_1), C_2(q_2))$ and the other contact information (e.g., normal vectors and contact points) are differentiable with respect to q_1 and q_2, which allows us to use gradient-based optimization techniques.

C. Markov Decision Process Formulation

Nevertheless, solving the optimization problem in Eq. (1) is extremely challenging due to the high-dimensional search space as the number of the objects n increases and the presence of complex physical constraints that involve non-convex and non-differentiable functions.

To address the problem Eq. (1) in a tractable manner, we model the problem as a sequential decision-making process where we place one object at a time. Suppose that we have already placed k objects in the scene and denote the set of indices of the placed objects as $\mathcal{P}_k \subset \{1, \cdots, n\}$. Then, the k-th scene $\mathcal{X}_k := \{(i, q_i) \mid i \in \mathcal{P}_k\}$ is the collection of object indices and their corresponding placement poses that have been placed in the scene. Also, let the k-th inventory $\mathcal{I}_k := \{1, \cdots, n\} \backslash \mathcal{P}_k$ be the set of remaining object indices that have not been placed yet. Given $(\mathcal{X}_k, \mathcal{I}_k, T)$, we now need to select an index $i_{k+1} \in \mathcal{I}_k$ and find the corresponding placement pose q_{k+1} for the object. The resulting scene is subsequently simulated to determine whether it is physically feasible and how well it approximates the target shape T.

The sequential decision-making process can be further modeled as a Markov Decision Process (MDP). We define the state space $\mathcal{S}$ as the set of states $s_k := (\mathcal{X}_k, \mathcal{I}_k, T)$, the action space $\mathcal{A}$ as the set of actions $a_k := (i_{k+1}, q_{k+1})$ representing the object index i_{k+1} and its placement pose q_{k+1}, the transition function $p : \mathcal{S} \times \mathcal{A} \to \mathcal{S}$ that maps a state-action pair to the next state, and the reward function $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ that evaluates the improvement in approximating the target shape T while satisfying the physical constraints.

More specifically, the pose q_{k+1} in an action is first inserted into the current scene to form a provisional scene $\tilde{\mathcal{X}}_{k+1} := \mathcal{X}_k \cup \{(i_{k+1}, q_{k+1})\}$. Then, the transition operator evaluates the physical simulation of the provisional scene under gravity and contact dynamics and returns a settled scene $\mathcal{X}_{k+1}$. Therefore, the transition function p can be defined as

$$s_{k+1} = p(s_k, a_k) = (\mathcal{X}_{k+1}, \mathcal{I}_{k+1}, T)$$

where $\mathcal{I}_{k+1} = \mathcal{I}_k \backslash \{i_{k+1}\}$. The implementation of this transition and the reward design are described in Sec. V. Therefore, the MDP can be defined as a tuple:

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, p, r, \gamma)$$

where $\gamma \in [0, 1]$ is the discount factor that determines the importance of future rewards.

D. Overall Structure of the Sequence Planner

The overall structure of the sequence planner is illustrated in Fig. 1. The sequence planner uses Monte Carlo Tree Search (MCTS) to explore the space of possible stacking sequences. For each expansion, it first uses the initial pose sampler π_{init} to generate a candidate pose $\hat{q}_{k+1}$, and then uses the pose optimization module π_{opt} to refine it into a locally optimal pose q_{k+1}. The structural simulation module π_{simul} subsequently implements the physical transition by evaluating the entire provisional scene and producing the settled next state. The resulting validity and simulation statistics are used to determine whether the action is admitted to the search tree and to evaluate its reward.

This gives the following execution flow:

$$s_k \xrightarrow{\pi_{\text{init}}} \hat{q}_{k+1} \xrightarrow{\pi_{\text{opt}}} q_{k+1} \xrightarrow{\pi_{\text{simul}}} s_{k+1},\ r(s_k, a_k).$$

We describe pose optimization in Sec. IV and sequence planning, including structural simulation and reward design, in Sec. V.

IV. Pose Optimization

Before presenting the other modules of our proposed sequence planner, we first introduce the pose optimization module π_{opt} that refines the candidate pose $\hat{q}_{k+1}$ generated by the initial pose sampler π_{init} to find a locally optimal placement pose q_{k+1} that is physically feasible and stable.

A. Single Object Pose Optimization Problem

Given s_k and the candidate pose $\hat{q}_{k+1}$ generated by π_{init}, the pose optimization module π_{opt} solves the following problem to obtain a locally optimal pose q_{k+1}:

$$\begin{aligned} \min_{q} \quad & \left[f_{\text{geo}}(q) + \min_{\lambda} f_{\text{phys}}(q, \lambda) \right] \\ \text{s.t.} \quad & g_{\text{cone},j}(q, \lambda_j) \le 0, \quad j = 1, \cdots, n_c \end{aligned} \tag{2}$$

where $q \in \text{SE}(3)$ is the pose of the object and $\lambda = (\lambda_1, \cdots, \lambda_{n_c}) \in \mathbb{R}^{3n_c}$ is the vector of contact forces at the n_c contact points between the object and the scene. This formulation follows prior contact-aware optimization methods [9, 5].

Each term in the objective of (2) is designed to evaluate the quality of the configuration in terms of geometric and physical feasibility. Here, $f_{\text{geo}}(q)$ measures geometric quality, including target shape approximation and collision avoidance; its detailed formulation is provided in the appendix. The physical objective $f_{\text{phys}}(q, \lambda)$ is designed to evaluate whether the contact forces λ are physically feasible and contribute to a stable configuration. For indices $l, m \in \mathcal{P}_{k+1} := \mathcal{P}_k \cup \{i_{k+1}\}$ such that $l \neq m$, let $I_{l,m} \subset \{1, \cdots, n_c\}$ be the set of contact point indices associated with the objects O_l, O_m, and let $I_l := \bigcup_{m \in \mathcal{P}_{k+1} \setminus \{l\}} I_{l,m}$ be the set of contact point indices associated with the object O_l. Then, the physical objective $f_{\text{phys}}(q, \lambda)$ is defined as the following:

$$\begin{aligned} f_{\text{phys}}(q, \lambda) = w_{\text{eq}} & \sum_{l \in \mathcal{P}_{k+1}} \| J_{c,I_l}^{\top}(q) \lambda_{I_l} + \lambda_{\text{ext},l} \|^2 \\ & + w_{\text{cp}} \sum_{j \in I_{i_{k+1}}} \mathrm{d}_j(q)^2 \|\lambda_j\|^2 + \rho \sum_{j=1}^{n_c} \|\lambda_j\|^2 \end{aligned} \tag{3}$$

The three terms respectively penalize violations of static equilibrium, contact complementarity between the candidate and the current scene, and large contact forces through regularization. Here, J_{c,I_l} is the contact Jacobian for object O_l, $\lambda_{\text{ext},l}$ is its external wrench, λ_{I_l} is the vector of contact forces at the contact points in I_l, and $w_{\text{eq}}, w_{\text{cp}}, \rho > 0$ are weighting and regularization parameters.

Finally, each $g_{\text{cone},j}(\cdot)$ is the smoothed friction cone constraints for the j-th contact point defined by

$$g_{\text{cone},j}(q, \lambda_j) = \sqrt{\|J_{c,j}^t \lambda_j\|^2 + \epsilon} - \mu_j J_{c,j}^n \lambda_j \tag{4}$$

where $\epsilon > 0$ is a smoothing parameter, μ_j is the friction coefficient, and $J_{c,j}^t$ and $J_{c,j}^n$ are the tangential and normal components of the Jacobian of the contact constraint.

We note that, due to the smoothness of DSFs, the terms in the objective function dependent to q are formulated which allows us to use gradient-based optimization techniques to optimize the placement pose of the object $O_{i_{k+1}}$.

B. Upper-Level Problem Optimization on SE(3) Manifold

Denoting $\lambda^*(q)$ as the optimal λ of the lower-level problem for a given pose q, we define the force dependent objective function as $f_{\text{phys}}(q) := f_{\text{phys}}(q, \lambda^*(q))$. Then, using the *Envelope theorem* [4], we can compute the gradient of $f_{\text{phys}}(q)$ with respect to q as follows:

$$\frac{df_{\text{phys}}(q)}{dq} = \frac{\partial f_{\text{phys}}(q, \lambda^*)}{\partial q} + \sum_{j=1}^{n_c} \nu_j^* \frac{\partial g_{\text{cone},j}(q, \lambda_j^*)}{\partial q}$$

where ν_j^* is the optimal dual variable associated with the j-th friction cone constraint $g_{\text{cone},j}(q, \lambda_j^*) \le 0$. Using the above gradient, we optimize the placement pose q on SE(3) manifold based on the Riemannian trust-region method [1] combined with Riemannian BFGS Hessian approximation [6] to find a locally optimal placement pose $q_{i_{k+1}}$ that minimizes the objective function $f_{\text{geo}}(q) + f_{\text{phys}}(q)$ while satisfying the friction cone constraints $g_{\text{cone},j}(q, \lambda_j^*(q)) \le 0$. In addition, motivated by homotopy based optimization methods [], we designed the sequence of the weight parameters in $f_{\text{geo}}(q)$ and $f_{\text{phys}}(q)$ such as w_{eq}, w_{cp} and iteratively solve the bi-level optimization problem (2) to induce a better local minimum that satisfies the physical constraints and approximates the target shape T.

C. Lower-Level Problem Optimization with ADMM

For a fixed q, the lower-level problem is a convex optimization problem with respect to λ which can be efficiently solved. Moreover, exploiting the inherent structure of the lower-level problem, we can reformulate it to be separable with respect

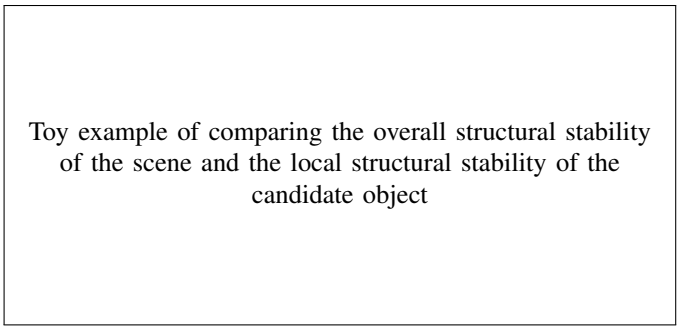


Fig. 2: Toy example of comparing the overall structural stability of the scene and the local structural stability of the candidate object.

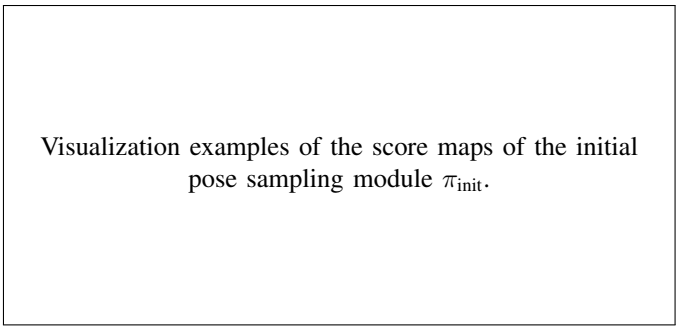


Fig. 3: Score maps of the initial pose sampling module π_{init}.

to the objects $\mathcal{P}_{k+1}$ in the scene and apply the alternating direction method of multipliers (ADMM) [3] to solve it in a scalable manner. For each contact force λ_j, we introduce two local copies $\lambda_{j,1}$ and $\lambda_{j,2}$ associated with the two objects in contact. Specifically, for $j \in I_{l,m}$ with $l < m$, $\lambda_{j,1}$ and $\lambda_{j,2}$ are associated with O_l and O_m, respectively. We then impose the consensus constraints

$$\lambda_{j,1} = \lambda_{j,2}.$$

Collecting all local copies into $\tilde{\lambda}$, the first term in $f_{\text{phys}}(q, \lambda)$ can be written as an objectwise-separable sum. The lower-level problem can therefore be reformulated as

$$\begin{aligned} \min_{\tilde{\lambda}, z} \quad & \tilde{f}_{\text{phys}}(q, \tilde{\lambda}) + \sum_{\substack{j=1,\dots,n_c \\ s=1,2}} \delta_{K_j}(z_{j,s}) \\ \text{s.t.} \quad & \lambda_{j,1} = \lambda_{j,2} = z_{j,1} = z_{j,2}, \qquad j = 1, \dots, n_c. \end{aligned}$$

where $\tilde{f}_{\text{phys}}(q, \tilde{\lambda})$ is the extended physical force function and δ_{K_j} is the indicator function of the second-order cone

$$K_j := \{\lambda \in \mathbb{R}^3 \mid g_{\text{cone},j}(q, \lambda) \leq 0\}.$$

Here, $z_{j,s}$ is an auxiliary variable associated with the j-th contact point and $s \in \{1, 2\}$. At each ADMM iteration, the objectwise-separable subproblem decomposes into $|\mathcal{P}_{k+1}|$ independent subproblems, one for each object O_l. The remaining updates consist of $2n_c$ projections onto the second-order cones K_j and $2n_c$ auxiliary-variable updates, which reduce to averaging. The projections can be computed in closed form. The detailed ADMM algorithm for the lower-level problem is provided in the appendix.

We want to emphasize that the force optimization approach considers the overall structural stability of the scene by evaluating the static equilibrium constraints for all objects in the scene.

V. Sequence Planning

In this section, we introduce the remaining modules π_{init} and π_{simul} that are used to generate candidate poses and simulate the stability of the scene, respectively. Then, we present the Monte Carlo Tree Search (MCTS) algorithm that uses these modules to explore the space of possible stacking sequences and find an optimal sequence of object placements.

A. Initial Pose Sampling

The initial pose sampler generates a candidate object, orientation, and planar translation from the current state $s_k = (\mathcal{X}_k, \mathcal{I}_k, T)$. Let $\mathcal{G}_{i,R}(s_k)$ denote the feasible planar sampling grid for object i and orientation R, and let $\hat{q}_{k+1} = (R, (\mathbf{x}, z_{i,R}(\mathbf{x}; s_k)))$. The sampler is defined by

$$\begin{aligned} &\pi_{\text{init}}(i, R, \mathbf{x} \mid s_k) = \pi_{xy}(\mathbf{x} \mid i, R, s_k)\pi_{\text{unif}}(i, R \mid s_k), \\ &\pi_{xy}(\mathbf{x} \mid i, R, s_k) \propto \exp(\sigma(\mathbf{x} \mid i, R, s_k)/\tau)\,. \end{aligned}$$

where $\pi_{\text{unif}}(i, R \mid s_k)$ is uniform over $i \in \mathcal{I}_k$ and the finite rotation set $\mathcal{R}$. The distribution π_{xy} is defined over $\mathbf{x} \in \mathcal{G}_{i,R}(s_k)$, and $\tau > 0$ controls the preference for higher-scoring planar locations. The initial height $z_{i,R}$ is computed from the scene geometry, while the full pose is subsequently refined by the pose optimization module π_{opt}. For brevity, the dependence of the score on s_k is omitted below. The proposal score $\sigma(\mathbf{x} \mid i_{k+1}, R)$ is formulated as a weighted sum of several geometric features that are computed based on the scene $\mathcal{X}_k$, the target shape T, and the object i_{k+1}:

$$\begin{aligned} \sigma(\mathbf{x} \mid i_{k+1}, R) = w_{\text{s}}\sigma_{\text{s}}(\mathbf{x} \mid i_{k+1}, R) + w_{\text{f}}\sigma_{\text{f}}(\mathbf{x} \mid i_{k+1}, R) \\ + w_{\text{b}}\sigma_{\text{b}}(\mathbf{x} \mid i_{k+1}, R) - w_{\text{h}}\sigma_{\text{h}}(\mathbf{x} \mid i_{k+1}, R), \end{aligned}$$

where σ_{s} is the support area score map, σ_{f} is the filled floor area score map, σ_{b} is the boundary proximity score map, and σ_{h} is the height score map, respectively with weight parameters w_{h}, w_{s}, $w_{\text{f}}, w_{\text{b}} > 0$.

Intuitively, the score σ_{s} encourages the object to be placed on a larger support area, the score σ_{f} encourages the object to fill the empty floor area, the score σ_{b} encourages to fill the boundary of the target shape T first, and the last score σ_{h} encourages the object to be placed at a lower height to improve stability. The visualization examples of the score maps are shown in Fig. 3 and the details of the score map computation will be described in the appendix.

B. Simulation Module

As mentioned in Sec. III, after π_{opt} refines the candidate pose $\hat{q}_{k+1}$ to find a locally optimal placement pose q_{k+1}, the simulation module π_{simul} acts as a transition function $s_{k+1} = p(s_k, a_k)$ which simulates the physical behavior of the scene $\mathcal{X}_{k+1}$ after placing the object $O_{i_{k+1}}$ at the placement pose q_{k+1}. The simulation module π_{simul} works in two steps: first,

Algorithm 1 Initial pose sampling

Require: State $s_k = (\mathcal{X}_k, \mathcal{I}_k, T)$, rotation set $\mathcal{R}$
1: Sample $i_{k+1} \sim \text{Unif}(\mathcal{I}_k)$ and $R \sim \text{Unif}(\mathcal{R})$
2: Construct the feasible planar grid $\mathcal{G}_{i_{k+1},R}(s_k)$
3: Evaluate $\sigma(\mathbf{x} \mid i_{k+1}, R, s_k)$ for all $\mathbf{x} \in \mathcal{G}_{i_{k+1},R}(s_k)$
4: Sample $\mathbf{x}$ from $\pi_{xy}(\mathbf{x} \mid i_{k+1}, R)$
5: Set $\hat{q}_{k+1} = (R, (\mathbf{x}, z_{i_{k+1},R}(\mathbf{x}; s_k)))$
6: **return** $(i_{k+1}, \hat{q}_{k+1})$

freezing the objects in the scene $\mathcal{X}_k$ and simulate the newly placed object $O_{i_{k+1}}$ to further refine the pose, and second, simulate the entire scene $\mathcal{X}_{k+1}$ to ensure that the entire scene is stable. For both steps, we use the early-termination condition to terminate the simulation if the scene is stable or expected to be unstable, and the simulation statistics are used to evaluate the reward of the action a_k. The early-termination condition is based on the mass-normalized kinetic energy $E_{\text{kin},i}$ and the accumulated energy $E_{\text{acc},i}$ of the object O_i in the scene $\mathcal{X}_{k+1}$, which are defined as

$$E_{\text{kin},i}(t) = \frac{1}{2m_i} \mathcal{V}_i^\top M_i \mathcal{V}_i, \quad E_{\text{acc},i}(t) = \sum_{s=0}^{t} \sqrt{E_{\text{kin}(s),i}} \Delta t$$

where $\mathcal{V}_i \in \mathbb{R}^6$ is the linear and angular velocity of the object O_i, $M_i \in \mathbb{R}^{6\times 6}$ is the mass matrix of the object O_i, m_i is the mass of the object O_i, and Δt is the simulation time step. Then, the early-termination condition is defined as follows:

$$\begin{aligned} &E_{\text{kin},i}(t) < \epsilon_{\text{kin}} \quad \text{for all } i \in \mathcal{P}_{k+1} \text{ or} \\ &E_{\text{acc},i}(t) > \epsilon_{\text{acc}} \quad \text{for some } i \in \mathcal{P}_{k+1} \end{aligned} \tag{5}$$

where $\epsilon_{\text{kin}}, \epsilon_{\text{acc}} > 0$ are the thresholds for the kinetic energy and the accumulated energy, respectively.

The simulation module is implemented using the GPU-accelerated subADMM algorithm [7] which is a scalable and efficient physics simulation algorithm for large-scale rigid body systems with contact.

C. Reward Design

The reward function is designed to encourage the agent to place objects in a way that maximizes the filled volume of the target shape T while ensuring the stability of the scene and formulated as follows:

$$\begin{aligned} r(s_k, a_k) = w_T \frac{\text{v}\left(\overline{\mathcal{X}}_{k+1}\right) - \text{v}\left(\overline{\mathcal{X}}_k\right)}{\text{v}(T)} + w_O \frac{\text{v}\left(O_{i_{k+1}} \cap T\right)}{\text{v}\left(O_{i_{k+1}}\right)} \\ - w_H \left(\max_{\mathbf{x}} H_{\mathcal{X}_{k+1}}(\mathbf{x}) - \mathbb{E}_{\mathbf{x}}[H_{\mathcal{X}_{k+1}}(\mathbf{x})] \right) \\ - w_D \mathbb{E}_{\mathcal{V}\sim\mathcal{N}(0,\Sigma)} \left[\delta q_{k+1}(\mathcal{V})\right] - w_S f_{\text{eq}} \end{aligned}$$

where $w_T, w_O, w_H, w_S, w_D > 0$ are the weight parameters for each reward term and $\overline{\mathcal{X}}_k := T \cap \bigcup_{i\in\mathcal{P}_k} O_i(q_i)$ is the intersection of the target shape T and the union of the placed objects in the scene $\mathcal{X}_k$. The first term and the second term encourage the agent to place objects that fill the target shape T and maximize the filled volume of the target shape T, respectively. The third term penalizes the height difference between the maximum height and the average height of the scene $\mathcal{X}_{k+1}$ to encourage a more uniform distribution of objects in the scene. The fourth term penalizes the expected pose deviation $\delta q_{k+1}(\mathcal{V})$ of the newly placed object $O_{i_{k+1}}$ under random perturbations $\mathcal{V}$ sampled from a Gaussian distribution $\mathcal{N}(0, \Sigma)$ to encourage the agent to place objects in a more stable manner. The last term penalizes the violation of the static equilibrium constraints f_{eq} from π_{opt} to encourage the agent to place objects in a physically feasible manner.

D. Monte Carlo Tree Search for Sequence Planning

Equipped with the modules π_{init}, π_{opt}, π_{simul} and the MDP $\mathcal{M}$, we now use Monte Carlo Tree Search (MCTS) to explore the space of possible stacking sequences. Given the root state $s_0 = (\mathcal{X}_0, \mathcal{I}_0, T)$, MCTS iteratively builds a search tree $\mathcal{T}$ where each node corresponds to a state s_k and each edge corresponds to an action a_k.

ACKNOWLEDGMENTS

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